

Understanding Technology Acceptance of Social Robots as Conversational Interfaces for LLMs

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Abstract—Advancements in large language models (LLMs) have transformed human-computer interaction, enhancing engagement with conversational agents like chatbots, virtual agents, and social robots. This study explored how different conversational interfaces (computer, Furhat, NAO, and Pepper) impact perception and acceptance using the Human-Robot Interaction Evaluation Scale (HRIES) and the Technology Acceptance Model (TAM). One hundred participants rated interfaces through a video-based survey, revealing significant differences: computers scored highest in Perceived Usefulness (PU) and Ease of Use (PEU), while social robots, particularly Pepper and NAO, excelled in Sociability. A Structural Equation Model (SEM) indicated that HRIES dimensions of Sociability and Agency positively influenced Perceived Enjoyment (PENJ) and PEU, whereas Disturbance negatively affected both. These findings highlight the nuanced interplay between interface design and user acceptance, suggesting that while factors like Sociability and Agency are pivotal for enhancing enjoyment and ease of use, they alone are insufficient to account for overall acceptance.

Keywords—*Technology Acceptance Model, Conversational Robots in Education, GenAI, ChatGPT*

I. INTRODUCTION

Despite significant research interest, social robots have struggled with adoption in the market [1]. The struggle arises from inflated expectations, high development costs, technological limitations, and a gap between research advancements and practical usability, leading to social robots failing to meet market demands and resulting in abandoned business ventures [2]. LLMs such as ChatGPT, on the other hand, have seen instant widespread adoption, with half of the workforce reporting its use [3]. By examining the relationships between the HRIES dimensions and TAM constructs in relation to students' acceptance of the different conversational interfaces for ChatGPT, this research aims to contribute to understanding the factors influencing the adoption of different forms of LLMs-powered social robots compared to standard devices such as notebooks in education.

Together with the release of Whisper in 2022, which is an automatic, multilingual speech recognition (ASR) system, ChatGPT can be used not only to generate human-like text but also for voice-based interaction by converting what the user says into text (speech-to-text), feeding it to the chatbot, and then converting ChatGPT's responses into spoken words (text-to-speech) [4]. This "leap beyond text" [5] has the potential to enhance effective communication with humanoid robots and, in this way, overcome a significant technological bottleneck of conversing with robots, which in the past hindered the full realization of the potential benefits offered by robotic systems

[6]. Nevertheless, while AI-powered social robots are designed to serve as social interaction partners, user perceptions of social robots may influence perceived usefulness and, consequently, their acceptance [7].

The Technology Acceptance Model (TAM) [8], [9] is a well-established framework for predicting technology adoption. In the TAM, Behavioral Intention (BI), which refers to the intention to use technology, is influenced by two key attitudinal predictors: Perceived Usefulness (PU) and Perceived Ease of Use (PEU). PU is defined as "the degree to which a person believes that using a particular system would enhance his or her job performance," while PEU is defined as "the degree to which a person believes that using a particular system would be free of effort" [8], [9]. Specifically, PU reflects the technology's perceived ability to improve task performance, and PEU reflects the perceived effortlessness of its use. Additionally, Perceived Enjoyment (PENJ), or the perception that a system is inherently enjoyable, also plays a role in influencing technology acceptance [10]. The TAM has been successfully used in previous research on the acceptance of social robots, e.g., in a workplace environment [11], in hotel settings [12], and in the context of the psychosocial well-being of older adults [13].

Furthermore, as [14] highlights, the acceptance of robotic interfaces for ChatGPT is influenced not solely by their functional capabilities, but even more so by users' perceptions of factors such as friendliness, trust, and overall appeal. These attributes significantly shape user interactions and engagement [14]. In general, individual variations in perceptions of social robots may be attributed to a wide range of factors, including the robot's morphology, actions, movement, understanding of the robots' capabilities, individual characteristics of users, including previous experience as well as a misalignment between human expectations and robot performance [6], [15], [16], [17], [18].

This study aimed to investigate how different types of conversational interfaces, i.e., computer/notebooks vs. Popular social robots such as NAO, Pepper, and Furhat, as interfaces for ChatGPT, may influence technology perception and acceptance. Specifically, the study aimed to explore possible differences of these interfaces in relation to the key constructs of the Technology Acceptance Model (TAM). Furthermore this study aimed to examine how the constructs of the Human-Robot Interaction Evaluation Scale (HRIES) shape users' acceptance of and interactions with these conversational interfaces. The research was guided by the following questions:

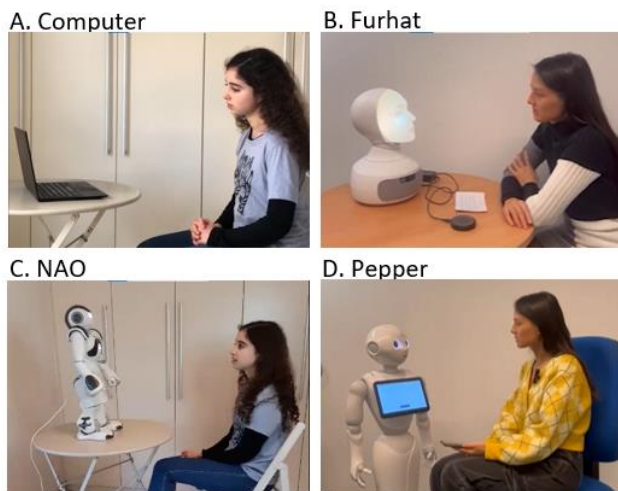


Fig. 1. The conditions: A. Computer, B. Furhat, C. NAO, and D. Pepper

RQ1. How does the type of conversational interface (computer vs. social robots: Furhat, NAO, Pepper) integrated with ChatGPT influence user perceptions and technology acceptance of these interfaces?

RQ2. How do the dimensions of the Human-Robot Interaction Evaluation Scale (HRIES) affect the key constructs of the Technology Acceptance Model (TAM) in relation to the different conversational interfaces for ChatGPT?

Our study hypothesized that users' perceptions of conversational interfaces for LLMs such as ChatGPT would influence perceived usefulness and ease of use and, consequently, intention to use these technologies as interfaces for ChatGPT. We created a hypothesized model that incorporates the HRIES constructs and prior experience with ChatGPT as possible factors influencing technology acceptance in relation to the TAM constructs (see Figure 4).

II. METHOD

A. Participants

One hundred participants (56 males, 43 females, 1 non-binary, age mean=27.44, SD=7.47, 63 USA and 37 German residents) completed a survey on the Prolific platform, an online service for data collection. The participants were randomly assigned to one of 4 conditions. They signed a consent form that was approved by the institutional ethical committee and received typical compensation for their participation.

B. Materials

The questionnaire was created using Qualtrics, an online survey platform, and was distributed and administered via Prolific, an online research platform designed to facilitate the recruitment and compensation of participants for academic studies. Prolific's filtering options were used to recruit participants who are currently students. We selected students since each of the four conditions included a video recording of a student chatting with ChatGPT via one of the four conversational interfaces. The participants were randomly assigned to watch one of four one-minute videos showcasing a voice interaction between a student and ChatGPT via a specific interface (see Fig 1).

The four videos featured the following interfaces for ChatGPT: (1) a computer (notebook, see Fig 2.A) ; (2) *Furhat* - a stationary social robot with a 3D-projected expressive face created by Furhat Robotics (see Fig 2.B); (3) *Naio* - a 58 cm bipedal humanoid robot equipped with cameras, microphones, and tactile sensors, developed by SoftBank Robotics (see Fig 2.C); and (4) *Pepper* - a 120 cm humanoid robot designed for social interaction, also manufactured by SoftBank Robotics, with a tablet interface, advanced facial recognition, and expressive gestures, enabling natural and intuitive human-robot communication (see Fig 2.D). ChatGPT 4 was used to generate the script for the recorded interactions. Two interactions were recorded with an 18-year-old student (Computer and NAO) and two with a 20-year-old student (Furhat and Pepper).

C. Conditions

The participants were randomly assigned to one of four conditions that differed only in the video presented at the beginning of the survey. All videos depicted the same scripted interaction between a student actor and either a computer or a robot. The only difference between the conditions was the type of communication interface: (1) a computer, (2) the robot Furhat, (3) the robot NAO, or (4) the robot Pepper. In all the videos, the voice-based interaction starts with the question from the interface: "What is your question about?". The student replies, "I have an exam next week, can you help me stay focused?". The conversation then proceeds with an answer generated by ChatGPT-4. The survey items were identical in all conditions, but the wording was adjusted, i.e., in the computer condition, the items referred to "the technology you saw in the video", while in the three robot conditions the items referred to "the robot you saw in the video".

D. Measures

1) Human-Robot Interaction Evaluation Scale (HRIES)

We applied the Human-Robot Interaction Evaluation Scale (HRIES) [19] to evaluate how study participants perceived the computer and the social robots (Pepper, NAO, and Furhat) as interfaces for ChatGPT, as presented in the videos. HRIES is a composite questionnaire with four dimensions: Sociability, Agency, Animacy, and Disturbance, each comprising a set of four semantic items. Factor analysis of HRIES demonstrated strong construct validity, with each dimension explaining distinct measures of users' social and psychological responses to robots [19]. The participants were asked to rate the HRIES items in relation to the interface they saw in the video on a 16-item Likert-type scale from 1 = not at all to 5 = totally. The internal consistency of the HRIES constructs was mostly high with (a) Sociability (4 items, $\alpha = .823$, e.g., "Likeable"), (b) Animacy (4 items, $\alpha = .686$, e.g., "Human-like"), (c) Agency (4 items, $\alpha = .726$, e.g., "Intentional"), (d) Disturbance (4 items, $\alpha = .856$, e.g., "Creepy").

2) Technology Acceptance Model (TAM) Questionnaire

A 17-item Likert-type scale (from 1=strongly disagree to 5=strongly agree) adjusted from [8] was used to measure the participants' attitudes towards the interface presented in the video. The internal consistency of the TAM constructs was high

with (a) *Perceived Usefulness (PU)* (6 items, $\alpha = .968$, e.g., “Using this technology in my studies would increase my productivity”), (b) *Perceived Ease of Use (PEU)* (6 items, $\alpha = .852$, e.g., “I would find it easy to get this technology to do what I want it to do.”), (c) *Behavior Intention (BI)* (2 items, $\alpha = .942$, e.g., “Assuming I had access to this technology, I intend to use it.”), and (d) *Enjoyment* (3 items, $\alpha = .815$, e.g., “I would have fun using this technology.”).

3) Demographics

Demographic data was collected, including gender, age, prior experience with technology, including personal computers, cleaning robots, social robots, and conversational AI chatbots (like chatGPT), and self-reported understanding of how conversational AI works.

E. Procedure

The participants chose to participate in this study by selecting it in the *Prolific* platform. The participants completed a questionnaire, which included the following sections: (1) Reading and signing a consent form, (2) Watching randomly assigned video, (3) Rating Technology Acceptance Model (TAM) items, (4) Rating Human-Robot Interaction Evaluation Scale (HRIES) items, (5) Providing demographics information. Overall, the survey took about 10 minutes to complete.

III. RESULTS

A. Technology Acceptance Model (TAM)

One-way MANOVA of Technology Acceptance Model (TAM) four constructs indicated a significant difference between the four conversational interfaces (computer, Furhat, NAO, and Pepper) Wilk's Lambda = .650, $F(3,100)=3.637$, $p<.0001$, $\eta^2 = .134$ (see Fig. 2). The Perceived Usefulness (PU) results revealed a significant effect of interface type on PU, $F(3, 100) = 4.627$, $p<.01$, $\eta^2 = .126$. Overall, PU scores were highest in the computer condition, followed by Pepper, NAO, and Furhat. A Tukey post-hoc test revealed that PU scores in the computer condition were significantly higher than in Furhat ($p<.01$) and NAO ($p<.05$) conditions (see Fig. 3.A).

Similarly, the Perceived Ease of Use (PEU) results revealed a significant effect of interface type on PEU $F(3, 100) = 4.063$, $p < .01$. Overall, PEU scores were highest in the computer condition, followed by Pepper, NAO, and Furhat. A Tukey post-hoc test revealed that the PEU scores in the computer condition were significantly higher than Furhat ($p<.01$) (see Fig. 3.B).

The Perceived Enjoyment (PENJ) results also revealed a significant effect of interface type on PENJ $F(3, 100) = 5.201$, $p < .01$, $\eta^2 = .140$. Overall, PENJ scores were highest in the Pepper condition, followed by the computer, NAO, and Furhat conditions. A Tukey post-hoc test revealed that PENJ scores in the Furhat condition were significantly lower than in the Pepper ($p<.01$), computer ($p<.01$), and NAO ($p<.01$) conditions (see Fig. 3.D).

Lastly, although the Behavioral Intentions (BI) scores in the computer condition were highest, followed by Pepper, NAO, and Furhat conditions, the differences between the interfaces in this category were not statistically significant (see Fig. 3.C).

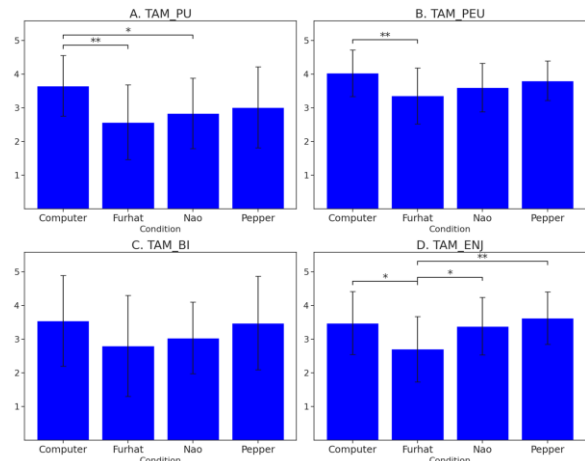


Fig. 2. Means and SDs of the Technology Acceptance Model (TAM) scale. * $p<.05$, ** $p<.01$

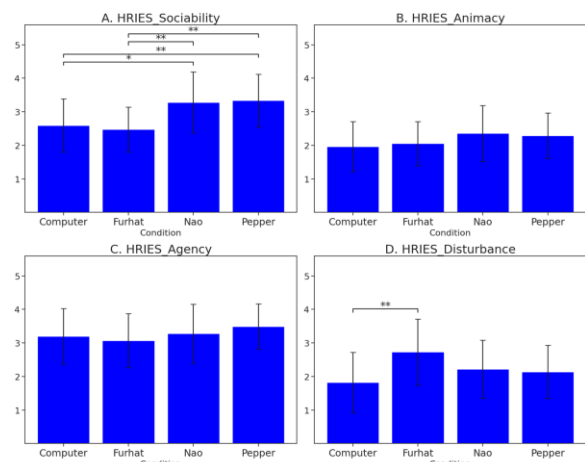


Fig. 3. Means and SDs of the Human-Robot Interaction Evaluation Scale (HRIES). * $p<.05$, ** $p<.01$

B. Human-Robot Interaction Evaluation Scale (HRIES)

One-way MANOVA of Human-Robot Interaction Evaluation Scale (HRIES) four constructs indicated a significant difference between the conversational interfaces (computer, Furhat, NAO, Pepper) Wilk's Lambda = .649, $F(3,100)=3.646$, $p<.0001$, $\eta^2 = .134$ (see Fig. 3).

The HRIES results revealed a significant effect of interface type on perceived Sociability, $F(3, 100) = 8.021$, $p<.0001$, $\eta^2 = .200$, with Pepper and NAO perceived as the most sociable interfaces, followed by the computer and the Furhat robot. A Tukey post-hoc test revealed that, indeed, Pepper was rated significantly more sociable compared to both the computer ($p<.01$) and the Furhat robot ($p<.01$). Likewise, NAO was also rated significantly more sociable compared to both the computer ($p<.05$) and Furhat ($p<.01$) (see Fig. 4.A).

In addition, the HRIES results revealed a significant effect of interface type on Disturbance, $F(3, 100) = 4.509$, $p<.01$, $\eta^2 = .123$. Furhat was perceived as most disturbing, particularly in the “spooky” and “scary” dimensions. A Tukey post-hoc test revealed that Furhat was rated significantly more disturbing than the computer ($p<.01$) but not significantly more disturbing

than the other robots (see Fig. 4.D). There were no significant differences between the interfaces regarding Animacy or Agency (see Fig. 4.B, 4.C).

C. Structural Equation Model (SEM) of HRIES and TAM

The structural equation model (SEM) was used to examine the relationships between constructs from the Human-Robot Interaction Evaluation Scale (HRIES) and the Technology Acceptance Model (TAM). The hypothesized structural equation model demonstrated an acceptable-good fit to the data. The Chi-square test was not significant, indicating a good model fit, $\chi^2(21) = 29.07$, $p = 0.11$. The Comparative Fit Index (CFI) was 0.98, also indicating a good fit. Similarly, the Goodness of Fit Index (GFI) was 0.94, and the Tucker-Lewis Index (TLI) was 0.96, both indicating a good fit. However, the Adjusted Goodness of Fit Index (AGFI) was 0.879, falling slightly below the good fit threshold, and the Root Mean Square Error of Approximation (RMSEA) was 0.062, which also falls in the acceptable range (0.05–0.08), suggesting that while the model is adequate, minor areas of misfit may exist (see Fig 4).

Perceived Usefulness (PU) emerged as the strongest predictor of Behavioral Intention to Use (BI) ($\beta = 0.816$, $p < .001$). Additionally, Perceived Ease of Use (PEU) exhibited a direct, though smaller, significant positive effect on BI ($\beta = 0.283$, $p < .05$). There was no direct effect between PEU and PU. Perceived Enjoyment (PENJ) had a strong and positive influence on both PU ($\beta = 0.689$, $p < .001$) and PEU ($\beta = 0.392$, $p < .001$).

Several relationships emerged between the four HRIES constructs and the TAM constructs. Disturbance had a direct, significant negative effect on PENJ ($\beta = -.399$, $p < .001$) and PEU ($\beta = -.238$, $p < .001$), but only had an indirect effect on PU. Sociability had a direct significant effect on PENJ ($\beta = 0.298$, $p < .001$) but only an indirect effect on PEU and PU. Agency had a direct, significant effect on PENJ ($\beta = 0.215$, $p < .05$) and PEU ($\beta = 0.165$, $p < .05$) but only an indirect effect on PU. No significant effects were found for Animacy. Prior experience of using chatGPT had a direct significant effect on PENJ ($\beta = -.191$, $p < .01$) but only an indirect effect on PEU and PU.

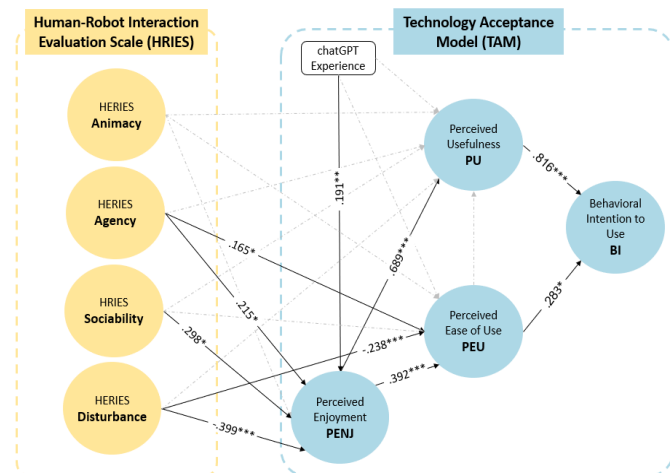


Fig 4. SEM of TAM and HRIES relationships. * $p < .05$, ** $p < .01$, *** $p < .001$

V. DISCUSSION

This study aimed to explore the impact of different types of conversational interfaces for LLMs, such as ChatGPT. We compared a computer interface to robotic interfaces (i.e., Furhat, NAO, and Pepper) in regard to the perception and acceptance of these four different technologies using the Human-Robot Interaction Evaluation Scale (HRIES) and the Technology Acceptance Model (TAM). Furthermore, the study examined the influence of the HRIES constructs on the TAM constructs.

Despite Pepper's and NAO's significantly higher ratings on Sociability compared to the computer and Furhat, the computer interface scored highest in terms of Perceived Usefulness (PU) and Perceived Ease of Use (PEU), followed by the robots Pepper, NAO, and Furhat, with significant differences between the computer and the robot Furhat. While the introduction of LLM like ChatGPT combined with social robots holds the potential to enhance the overall user experience [6], these findings align with prior research indicating that users are (currently) more familiar and comfortable with traditional computing devices, which often results in higher usability ratings [8], [9].

These results could be partially explained by the computer being rated lowest on Disturbance, significantly lower than the robot Furhat, which was also rated lowest on Perceived Enjoyment (PENJ). Indeed, the structural equation model (SEM) analysis indicated that while Sociability had a significant positive influence on PENJ and PEU, Disturbance had a significant negative influence on PENJ and PEU, with corresponding indirect influences on PU through PENJ.

Furhat's low ratings on Sociability and PENJ and high ratings on Disturbance could be attributed to the "uncanny valley" phenomenon, where highly anthropomorphic yet not entirely human-like features create discomfort among users [20][21]. Furhat's lack of a body, combined with its projected facial expressions, may have contributed to these results.

Several limitations of this study should be noted. Since this study was a collaboration between two labs from different continents, it necessitated local recordings, where each participating lab created two videos with a different actress, which could have impacted the perception of these videos. Nevertheless, one-way ANOVAs showed no significant differences based on the actress, indicating that this did not affect our findings. In addition, in this study, the participants rated a video and not an actual interaction they experienced themselves, which was shown to impact perceptions such as uncanny [22]. Future research should examine real-world interactions with emerging robotic interfaces.

In conclusion, this study highlights the nuanced interplay between the type of conversational interfaces for LLMs and user perceptions of their usability, enjoyment, and overall acceptance. While robots like Pepper and NAO excelled in Sociability and Enjoyment, the computer interface excelled in Usefulness and Ease of Use. However, these results may evolve as social robots become more prevalent and seamlessly integrated as interfaces for LLMs.

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